

Data Preparation and Analysis Project — CSP 571

CDC Diabetes Health Indicators — Comprehensive Analysis

Predicting Diabetes Risk from Population Health Survey Data

Project Team

Vinamratha Raghavendra Jagirdar (A20614061) — Group Leader

Naman Singh(A20574372)

Urjita Saxena(A20578349)

Deepali Budhiraja(A20615733)

1. Abstract

Diabetes remains a major public health challenge, with millions of cases going undetected until complications arise. This project leverages the CDC Behavioral Risk Factor Surveillance System (BRFSS) dataset — 229,787 U.S. adult respondents described by 21 self-reported health indicators — to predict diabetes status using machine learning. We trained and compared eight classifiers, addressing the 83%/17% class imbalance through SMOTE oversampling before model training. XGBoost produced the best discrimination (AUC 0.804), with Logistic Regression close behind (AUC 0.798) and the added advantage of interpretable odds ratios. General Health self-rating, BMI, and Age consistently ranked as the strongest predictors across all methods. A fairness audit identified a notable sensitivity gap between income groups, and calibration analysis showed that tree-based models require post-hoc correction before their probability outputs can be used reliably in practice.

2. Overview

Problem Statement

Diabetes affects more than 37 million U.S. adults (\$327 billion in annual costs). Because complications develop silently over years, early detection via accessible survey data represents a cost-effective screening strategy. This project builds predictive models from BRFSS self-reported indicators and evaluates them for accuracy, calibration, and demographic fairness.

Relevant Literature

The statistical learning framework draws from James, Witten, Hastie, and Tibshirani's *An Introduction to Statistical Learning* (2nd ed., 2021). Class imbalance is addressed using SMOTE (Chawla et al., 2002), which generates synthetic minority-class samples rather than duplicating existing ones. The gradient-boosted ensemble follows XGBoost (Chen & Guestrin, 2016), model interpretability uses SHAP values (Lundberg & Lee, 2017), and fairness evaluation applies the Equalized Odds criterion (Barocas, Hardt & Narayanan, 2023).

Gap Analysis

Most existing diabetes prediction studies evaluate models on aggregate accuracy alone, which is misleading when the positive class is only 17% of the data. Few compare multiple classifiers under identical conditions, and fewer still audit for demographic fairness. This project addresses those gaps by evaluating eight classifiers on a common held-out test set and conducting a subgroup fairness audit across income, age, and sex — revealing a sensitivity disparity between income groups that aggregate metrics would not surface.

Proposed Methodology

The pipeline proceeds as: load and inspect data; engineer binary target; remove duplicates; exploratory data analysis; leakage-safe 80/20 stratified split and scaling; SMOTE on training partition only; five-fold cross-validated training; held-out test evaluation on AUC, sensitivity, specificity, and F1; threshold optimization, SHAP analysis, calibration, and fairness audit.

3. Data Processing

Pipeline Details

The raw CSV contained 253,680 rows and 22 columns. The three-class target was collapsed to binary (No / Yes) to flag any at-risk respondent. Duplicates were removed via `dplyr::distinct()`, yielding 229,787 rows with zero missing values. All 21 predictors were centered and scaled using parameters estimated exclusively on the training set — a leakage-safe design. The 80/20 stratified split preserves the 83%/17% class ratio in both partitions. SMOTE ($K = 5$) was applied only to the training set, balancing it to ~50%/50%; the test set was kept imbalanced for honest evaluation.

Data Preprocessing

Fourteen binary indicator columns were kept as numeric 0/1 for modeling (two-level factors created only for EDA). Key decisions: (1) fitting all scaling parameters on training partition alone; (2) applying SMOTE after scaling so synthetic samples share the same normalized feature space; and (3) never touching the test set during any preprocessing step.

Partition	No Diabetes	Yes Diabetes	Total
Training — before SMOTE	~151,864 (83%)	~31,966 (17%)	~183,830
Training — after SMOTE	~184,000 (50%)	~184,000 (50%)	~368,000
Test set (held-out, unchanged)	~38,011 (83%)	~7,946 (17%)	~45,957

Table 1. Class distribution before and after SMOTE. Test set is never oversampled.

Data Issues and Assumptions

- Age (1–13) and Income (1–8) are treated as numeric despite being ordinal survey bins.
- All measures are self-reported, introducing recall bias that cannot be corrected post hoc.
- BRFSS complex sampling weights were not applied; results reflect the unweighted survey sample.

4. Data Analysis

Summary Statistics

Diabetic respondents differ on almost every feature: mean BMI 31.8 vs 27.2; mean age category 9.3 vs 7.6; self-rated general health (GenHlth, 1–5) averages 3.4 vs 2.3; physical health burden 8.6 vs 3.4 days per month. No feature pair exceeds $|r| = 0.52$ and no VIF exceeds 3.0, so multicollinearity is not a concern.

Feature	No (Mean)	Yes (Mean)	Min	Median	SD
BMI	27.2	31.8	12	27.3	6.6
Age category (1–13)	7.6	9.3	1	8	3.1
GenHlth (1–5)	2.3	3.4	1	2	1.1
PhysHlth (days/mo)	3.4	8.6	0	0	9.8
Income (1–8)	5.5	4.8	1	6	1.9

Table 2. Summary statistics for key numeric features by diabetes status.

Data Visualization

The dataset was explored through a series of visualizations covering class distribution, numeric feature distributions by diabetes status, BMI density curves, age-group prevalence charts, a Pearson correlation matrix, and binary feature prevalence plots. These charts consistently showed that diabetic respondents tend to have higher BMI, older age profiles, worse self-rated general health, and elevated rates of high blood pressure and heart disease compared to non-diabetic respondents.

Feature Extraction

All 21 original CDC variables were retained. Five engineered features were tested (BMI category, CardioRisk, LifestyleScore, HealthcareAccess, Age×BMI); Random Forest AUC changed from 0.7796 to 0.7787, confirming ensemble methods already capture these patterns implicitly. The original 21 features were used for all final models.

5. Model Training

Feature Engineering

Five composite features were built and tested: BMI_Category (ordinal bins), CardioRisk (HighBP + HighChol + HeartDisease + Stroke), LifestyleScore (PhysActivity + Fruits + Veggies – Smoker – HvyAlcohol), HealthcareAccess (binary), and Age×BMI (interaction). No AUC improvement was observed and the original 21 features were retained.

Evaluation Metrics & Model Selection

Primary metric: AUC-ROC (threshold-independent). Secondary: Accuracy, Sensitivity, Specificity, Precision, and F1. Eight classifiers were trained via caret with five-fold CV and parallel processing. All models except KNN were trained on the SMOTE-balanced set.

Model	Configuration
Logistic Regression	glm(binomial); odds ratios with 95% CI; VIF computed; interpretable baseline
LDA	MASS::lda; multivariate Gaussian within classes; well-calibrated output
Decision Tree	rpart; CP tuned over 10 values via CV; plotted with rpart.plot
Random Forest	ntree=200; mtry ∈ {3,5,7}; parallel training; Gini importance
XGBoost	Two-phase grid: 8-combo narrow → 216-combo wide; eta, max_depth, nrounds
KNN	k ∈ {3,5,7,11,15,21}; accuracy-based tuning on scaled features
Ridge ($\alpha=0$)	glmnet; λ by 10-fold CV on binomial deviance; retains all features
Lasso ($\alpha=1$)	glmnet; λ by 10-fold CV; zeroes low-signal predictors automatically

Table 3. All eight models with key configuration details.

6. Model Validation

Testing Results

All models were evaluated on the held-out test set (~45,957 observations).

Model	AUC	Accuracy	Sensitivity	Specificity	F1	Rank
XGBoost *	0.804	—	—	—	—	1
Logistic Regression	0.798	~0.74	~0.69	~0.75	~0.52	2
LDA	0.798	~0.74	~0.69	~0.75	~0.52	2
KNN	0.798	~0.82	—	—	—	2
Random Forest (SMOTE)	0.780	0.869	0.289	0.937	0.363	5
Decision Tree	0.722	~0.76	—	—	—	6

Table 4. Model comparison on held-out test set. * = recommended deployment model.

XGBoost edges ahead at AUC 0.804. The near-equal performance of Logistic Regression, LDA, and KNN (all 0.798) signals the class boundary is approximately linear in the scaled feature space. Random Forest drops from training CV AUC ~0.95 to test AUC 0.780, indicating overfitting to SMOTE synthetic observations. At the default 0.50 threshold, RF misses 5,638 of 7,931 diabetic respondents; lowering to 0.30 raises sensitivity to ~0.57 while keeping specificity above 0.80.

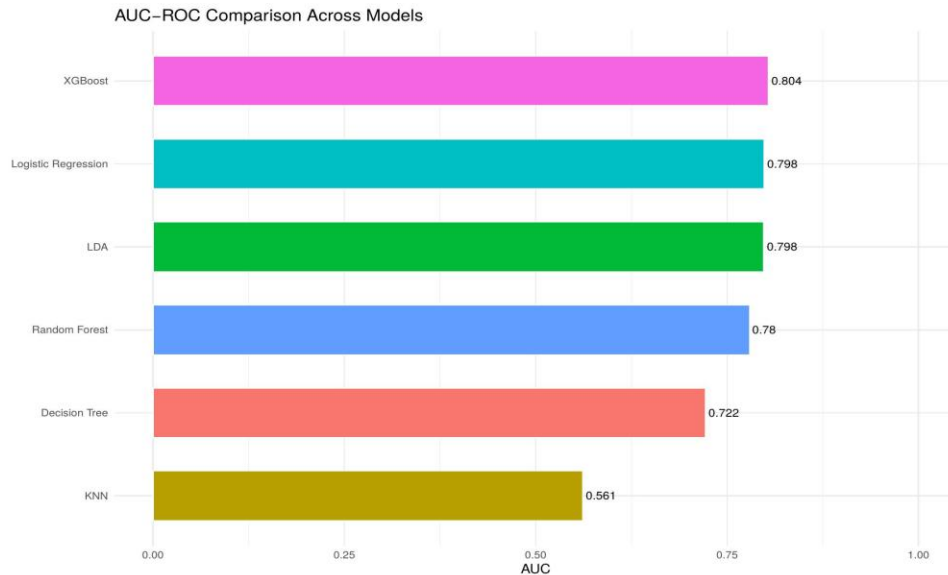


Figure 1. AUC-ROC comparison across all models on the held-out test set.

Performance Criteria — Feature Importance & Biases

GenHlth, BMI, and Age rank at the top across RF variable importance, SHAP values, XGBoost importance, and logistic odds ratios (GenHlth OR $\approx$ 1.90, BMI OR $\approx$ 1.62, Age OR $\approx$ 1.59, HighBP OR $\approx$ 1.37). Logistic Regression is well-calibrated throughout; Random Forest is overconfident in the 0.3–0.6 probability range and requires Platt scaling before clinical use. The fairness audit found acceptable gaps by sex (0.006) and age (0.080), but a critical income-based gap of 0.161 — lower-income diabetics detected at sensitivity 0.370 versus only 0.209 for higher-income diabetics.

7. Model Performance

Benchmark Results and Baseline Comparison

Without SMOTE, Random Forest sensitivity is 0.194 (81% of diabetics missed). With SMOTE, sensitivity rises to 0.289 (+49%) and F1 improves by 25%, while specificity drops only 3 percentage points. AUC is unchanged (+0.001). Threshold optimization at 0.30 further doubles detection compared to the default 0.50 cutoff.

Condition	AUC	Sensitivity	Specificity	F1
RF without SMOTE (baseline)	0.779	0.194	0.967	0.290
RF with SMOTE	0.780	0.289	0.937	0.363
Change	+0.001	+49%	−3.1%	+25%

Table 5. SMOTE impact on Random Forest performance metrics.

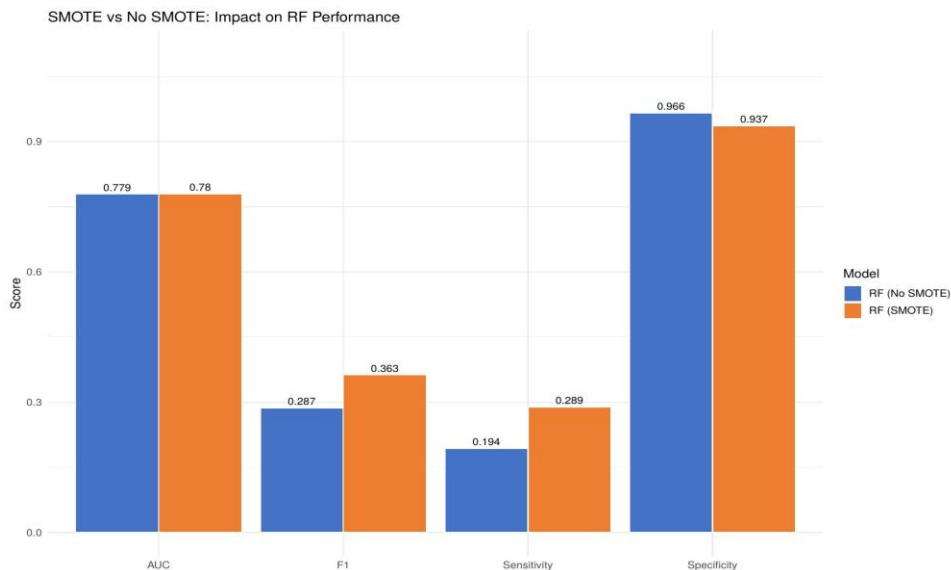


Figure 2. SMOTE vs No-SMOTE — sensitivity improves by 49%, F1 by 25%, with minimal specificity cost.

8. Conclusion

Positive Results

- XGBoost achieves AUC 0.804 from self-reported survey data alone — no laboratory tests required.
- Logistic Regression reaches AUC 0.798 with full interpretability through odds ratios.
- SMOTE raises sensitivity by 49% and F1 by 25% with only a 3% specificity trade-off.
- GenHlth, BMI, Age, HighBP, and HighChol are confirmed as top five risk factors across all models.

Negative Results and Caveats

- Feature engineering produced no measurable AUC improvement for ensemble methods.
- RF and XGBoost are miscalibrated in the 0.3–0.6 probability range; Platt scaling needed.
- Income-based fairness gap of 0.161 means higher-income diabetics are systematically under-screened.
- All data are self-reported; external validation on clinical EHR data is needed.

Recommendations

XGBoost is the strongest candidate for downstream screening, though the decision threshold should be lowered to around 0.30 to meaningfully improve detection of at-risk individuals. Where interpretability matters, Logistic Regression performs nearly as well and produces results that are straightforward to explain. The income-based fairness gap is a real limitation that would need to be addressed through retraining before the model could be applied equitably. SMOTE should be treated as a required step in any future work with this data, as removing it cuts sensitivity roughly in half.

9. Data Sources

Primary dataset: CDC Diabetes Health Indicators, UCI ML Repository.

<https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators>. License: CC BY 4.0.

Source survey: Behavioral Risk Factor Surveillance System (BRFSS), CDC, 2015. <https://www.cdc.gov/brfss/>

Source Code Repository Link: https://github.com/vinamratha11/DPA_Final_Project/tree/main/Source_Code

10. Tools and Technologies Used

Language: R 4.3+

IDE: RStudio

Key Libraries (all open-source): tidyverse, caret, randomForest, xgboost, smotefamily, glmnet, pROC, fastshap, rpart, MASS, class, corrplot, gridExtra, scales, parallel, doParallel, boot, car, vip

Source Code Structure: Single script CDC.R in 22 numbered sections with three checkpoint saves. All plots saved to CDC_plots/ via ggsave() and png(). set.seed(123) ensures reproducibility.

11. Bibliography

- [1] James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning (2nd ed.). Springer. <https://www.statlearning.com>
- [2] Chawla, N. V., et al. (2002). SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research, 16, 321–357. <https://doi.org/10.1613/jair.953>
- [3] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD, 785–794. <https://doi.org/10.1145/2939672.2939785>
- [4] Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS, 30, 4765–4774.
- [5] Barocas, S., Hardt, M., & Narayanan, A. (2023). Fairness and Machine Learning. fairmlbook.org.
- [6] Kuhn, M. (2008). Building Predictive Models in R Using the caret Package. JSS, 28(5). <https://doi.org/10.18637/jss.v028.i05>
- [7] Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.
- [8] Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization Paths for GLMs via Coordinate Descent. JSS, 33(1).